Praneet **Vayalali**

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Summary

An engineer with a strong passion for aviation and specializing in flight dynamics and control. My current focus is on developing the XR ground control station and the flight simulators, ensuring its comprehensive functionality for the certification of the XPro & VoloCity aircrafts.

Education

Ph.D. in Aeronautical Engineering — *Aug. 2016 - Dec. 2021*
Rensselaer Polytechnic Institute (RPI), **GPA - 3.83/4** — *Troy, NY, USA*

Master of Technology in Aerospace Engineering — *July 2014 - July 2016*
Indian Institute of Technology Kanpur (IIT K), **GPA - 9.25/10** — *Kanpur, India*

Bachelor of Technology in Aerospace Engineering — *July 2009 - June 2013*
Amrita School of Engineering (ASE), **GPA - 8.91/10** — *Coimbatore, India*

Technical Skills

Softwares	FLIGHTLAB, MATLAB/Simulink, CONDUIT, CIFER, LabView
Languages	C++, Python
Interfaces	Serial(UART / RS-232), RS-422/RS-485, CAN
General	Git, Siemens Teamcenter, X-Plane, Linux, Microsoft Office Suite, LaTeX

Work & Research Experience

Flight Simulation Engineer — *Dec. 2021 - Current*
Flight Science Department, Volocopter Technologies GmbH — *Bruchsal, Germany*

- Developed XR ground control station software for precision flight test point execution within a 4-person team, progressing from 1-month prototype to refined implementation.
- Developed engineering flight simulators for the VoloCity and XPro aircrafts, including flight dynamics modeling and SIL/HIL integration of avionics subsystems, across two platforms: fixed-base dome and mixed-reality motion-based simulators.
- Performed V&V of modeled and integrated systems (flight controls, displays, flight mechanics model, motor model, BMS) using test bench and flight test data.
- Provided test development and execution support to pilots and design engineers for handling qualities, flight controls, aircraft flight manual, safety, and human factors.
- Defined and implemented development processes in accordance with EASA MOC-2 SC-VTOL (MOC4 VTOL.2500(b)) for flight simulator.
- Planning and coordination of flight simulator development to support aircraft design development and certification process.
- Supported the preparation and review of simulator, handling qualities, and aeroelasticity certification documents (certification plans, analysis reports, test proposals/results).
- Performed predictive handling qualities assessment, aeroelastic and aeroservoelastic stability analysis for notch filter design and safety of flight evidence.

Post-Damage Reconfiguration for Rotorcraft with Control Redundancy — *Aug. 2016 - Oct. 2021*
Ph.D. Dissertation, Advisor - Dr. Farhan Gandhi — *RPI, NY, USA*

- Examined adaptive and robust fly-by-wire flight control strategies that can tolerate damage/degradation in various control effectors to enable safe flight on VTOL aircraft with control redundancy.
- Performed nonlinear flight simulation and predictive handling qualities (ADS-33E) analysis on the use of various control allocation and modern control design techniques on a UH-60 Black Hawk and derivative compound helicopter platforms.

Post-Failure Control Reconfiguration on a High-Speed Lift-Offset Coaxial Helicopter — *Aug. 2019 - Oct. 2021*
Research Project, Advisor - Dr. Farhan Gandhi — *RPI, NY, USA*

- Worked in a team of two to investigate the potential for fault compensation in flight for a lift-offset coaxial helicopter by examining the interchangeability of controls at different points within the flight envelope, thereby identifying allowable fault cases for the vehicle.
- Supported with the development of an elastic blade coaxial-pusher helicopter trim model based on the X2 Technology™ demonstrator which was then used to explore trim control variation in different flight regimes.

Frequency Domain Based System Identification of a Small-Sized RUAV — *May 2015 - Jul. 2016*
Master's Thesis, Advisor - Dr. A. Abhishek — *IIT Kanpur, India*

- Characterized a fully instrumented small-size rotary-wing unmanned aerial vehicle (RUAV) using a frequency domain based experimental system identification technique.
- Flight test of a 1 kg class RUAV was conducted and its responses were recorded. The closed-loop responses were analyzed to extract a bare-airframe 6-DOF linear flight dynamics state space model of the aircraft for hover flight condition. This was followed up by flight test validation.

Projects

Backstepping Control for a Variable-RPM Quadrotor Aircraft *Jan. 2019 - May 2019*
NONLINEAR CONTROL SYSTEMS COURSE PROJECT *RPI, NY, USA*

- Demonstrated the applicability of backstepping control design to a hobbyist scale quadrotor vehicle.
- Examined the performance of a backstepping feedback controller in comparison to a PID controller on the quadrotor.

Extreme Altitude Mountain Rescue Vehicle *Jan. 2019 - May 2019*
THE VERTICAL FLIGHT SOCIETY'S 36TH ANNUAL STUDENT DESIGN COMPETITION *RPI, NY, USA*

- Proposed a synchropter as a VTOL aircraft design capable of search and rescue missions at the summit of Mt Everest, where only one helicopter has ever landed and taken off again.
- Developed a flight dynamics model and analyzed the autonomous dynamics of the proposed rotorcraft design. Design constraints include freezing temperatures, thin air and hostile weather conditions with degraded visual environment all contribute to making rescue work in high-altitude environments particularly dangerous or impossible.
- Won third place in the graduate category the vertical flight society's 36th annual student design competition.

Power Optimization for UH-60 Black Hawk in Cruise using Variable RPM and Stabilator *Sept. 2017 - Dec. 2017*
INDEPENDENT STUDY FOR ADVANCED DESIGN OPTIMIZATION COURSE *RPI, NY, USA*

- Optimized power using the main rotor speed and stabilator pitch incidence as design variables on a physics-based UH-60 Black Hawk helicopter simulation model at a cruise speed of 140 knots.

Controller Design for a Quadrotor UAV *Jan. 2017 - May 2017*
HELICOPTER DESIGN COURSE PROJECT *RPI, NY, USA*

- Nonlinear flight simulation model with PID and LQR controllers were developed for a 2 kg quadrotor helicopter and their performance was compared in simulation.
- Compared control schemes using controller gains tuned for the hover flight condition and flown for single waypoint and multiple waypoint trajectories.

Design of V/STOL Air Taxi *Dec. 2014 - May 2015*
AIAA FOUNDATION GRADUATE TEAM AIRCRAFT DESIGN COMPETITION *IIT Kanpur, India*

- Worked with a team of seven to propose a design for an air taxi system that operates out of confined urban areas requiring vertical and short takeoff and landing capability.
- Involved in the design of a ducted rotor system for the V/STOL aircraft with consideration of noise and safety.

Journal Articles

- McKay, M., **Vayalali, P.**, Gandhi, F., Berger, T., Lopez, M.J.S., "Control Allocation Reconfiguration for Actuator Failure on a Coaxial-Pusher Helicopter", Journal of the American Helicopter Society, Vol. 68(3), July 2023, pp. 32004-32015(12), DOI:10.4050/JAHS.68.032004.
- **Vayalali, P.**, McKay, M., Krishnamurthi, J., Gandhi, F., "Fault-Tolerant Control on a UH-60 Black Hawk Helicopter using Horizontal Stabilator," CEAS Aeronautical Journal, Vol. 12(1), Jan. 2021, pp. 13-27, DOI:10.1007/s13272-020-00476-5.
- **Vayalali, P.**, McKay, M., Krishnamurthi, J., Gandhi, F., "Horizontal Stabilator Utilization for Post Swashplate Failure Operation on a UH-60 Black Hawk Helicopter," Journal of the American Helicopter Society, Vol. 65, Apr. 2020, pp. 1–13(13), DOI:10.4050/JAHS.65.022009.

Conference Proceedings

- **Vayalali, P.**, McKay, M., Gandhi, F., "Fault-Tolerant Control Allocation on a Compound Helicopter in Cruise," Proceedings of the Vertical Flight Society 77th Annual Forum, Virtual, 10-14 May 2021.
- McKay, M., **Vayalali, P.**, Gandhi, F., Berger, T., Lopez, M. J. S., "Redistributed Allocation for Flight Control Failure on a Coaxial Helicopter," Proceedings of the Vertical Flight Society 77th Annual Forum, Virtual, 10-14 May 2021.
- **Vayalali, P.**, McKay, M., Gandhi, F., "Redistributed Pseudoinverse Control Allocation for Actuator Failure on a Compound Helicopter," Proceedings of the Vertical Flight Society 76th Annual Forum, Virtual, 6-8 Oct 2020.
- McKay, M., **Vayalali, P.**, Gandhi, F., "Post-Failure Control Reconfiguration for a Lift-Offset Coaxial Helicopter," Proceedings of the Vertical Flight Society 76th Annual Forum, Virtual, 6-8 Oct 2020.
- **Vayalali, P.**, McKay, M., Krishnamurthi, J., Gandhi, F., "Robust Use of Horizontal Stabilator in Feedback Control on a UH-60 Black Hawk," Proceedings of the Vertical Flight Society 75th Annual Forum, Philadelphia, PA, 13-17 May 2019.
- **Vayalali, P.**, McKay, M., Krishnamurthi, J., Gandhi, F., "Swashplate Actuator Failure Compensation for UH-60 Black Hawk in Cruise Using Horizontal Stabilator," Proceedings of the 74th American Helicopter Society Annual Forum, Phoenix, AZ, 7-10 May 2018.

Honors & Awards

2018	**Third Place,** The Vertical Flight Society's 36th Annual Student Design Competition	*Virginia, USA*
2017	**Runner-up,** The American Helicopter Society Northeast Region Robert L. Lichten Award Competition	*Stratford, USA*
2012	**Second Place,** RC aircraft modeling event as a part of techfest of Amrita School of Engineering	*Coimbatore, India*
2010	**First Place,** SAE College level Aero Modeling Tier I event	*Coimbatore, India*